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**Department
of
Electronics and Communication Engineering**

**Minor Project Report
on
FPGA Based Real Time Wireless Communication
Framework.**

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CERTIFICATE

This is to certify that project entitled **FPGA Based Real Time Wireless Communication Framework** is a bonafide work carried out by the student team of **"Pooja R K - 01FE23BEC080, Pragati Abbigeri - 01FE23BEC089, Bhavani M G - 01FE23BEC134, Sahana Mattur - 01FE23BEC153 "**. The project report has been approved as it satisfies the requirements with respect to the Minor project work prescribed by the university curriculum for BE (VI semester) in Department of Electronics and Communication Engineering of KLE Technological University for the academic year 2025-26.

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-Pooja R K , Pragati Abbigeri , Bhavani M G, Sahana Mattur

ABSTRACT

Intelligent Transportation Systems (ITS) require reliable and real-time communication frameworks to improve traffic management, road safety, and information exchange between transportation entities. The increasing demand for smart transportation solutions has created the need for efficient communication systems capable of supporting low-latency and wireless data transmission. This project presents an FPGA-Based Real-Time Communication Framework for Intelligent Transportation Systems using the PYNQ-Z2 FPGA board, NRF24L01 wireless transceiver modules, and Arduino UNO. The proposed system consists of a transmitter section implemented using the PYNQ-Z2 FPGA platform interfaced with an NRF24L01 wireless transceiver through the SPI protocol. The FPGA processes input information and transmits it wirelessly using the 2.4 GHz ISM band. The receiver section consists of another NRF24L01 transceiver connected to an Arduino UNO, which receives the transmitted data and displays the received information through the serial monitor. The developed communication framework demonstrates reliable wireless communication between the transmitter and receiver nodes. The FPGA provides high-speed processing and real-time operation, while the NRF24L01 modules enable low-cost and efficient wireless communication. The successful transmission and reception of information validate the feasibility of implementing FPGA-based communication frameworks for Intelligent Transportation System applications. The proposed system can be extended for various ITS applications such as vehicle-to-infrastructure communication, traffic monitoring systems, emergency information dissemination, smart traffic management, and intelligent road safety solutions. The project highlights the effectiveness of combining FPGA technology and wireless communication to develop scalable and efficient communication frameworks for future transportation systems.

Keywords: Intelligent Transportation System (ITS), FPGA, PYNQ-Z2, NRF24L01, Arduino UNO, Wireless Communication, SPI Protocol, Real-Time Communication, Traffic Monitoring.

Contents

1	Introduction	6
1.1	Motivation	6
1.2	Intelligent Transportation Systems	6
1.3	FPGA in Transportation Applications	7
1.4	Wireless Communication in Intelligent Transportation Systems	7
1.5	Objectives	7
1.6	Literature Survey	8
1.7	Problem Statement	8
1.8	Organization of the Report	8
2	System Design	10
2.1	Introduction	10
2.2	Need for Real-Time Communication in Intelligent Transportation Systems	10
2.3	Overall System Architecture	11
2.4	Transmitter Section	12
2.4.1	Input Switches	12
2.4.2	PYNQ-Z2 FPGA Board	12
2.4.3	AXI GPIO	12
2.4.4	AXI SPI	12
2.4.5	NRF24L01 Transmitter	13
2.5	Receiver Section	13
2.5.1	NRF24L01 Receiver	13
2.5.2	Arduino UNO	13
2.5.3	Serial Monitor	13
2.6	Wireless Communication Channel	13
2.7	Communication Protocol	14
2.8	System Operation	14
2.9	Advantages of the Proposed System	14
2.10	Summary	15
3	Implementation Details	16
3.1	Specifications and System Architecture	16
3.2	Algorithm	17
3.3	PYNQ-Z2 FPGA Board	18
3.4	NRF24L01 Wireless Transceiver	18
3.5	Arduino UNO	18
3.6	Vivado Design Implementation	18
3.7	Flowchart	19

4	Results and Discussions	21
4.1	Results and Discussions	21
4.1.1	Vivado Block Design	21
4.1.2	Vitis Application Execution	22
4.1.3	Transmitter Hardware Setup	22
4.1.4	Receiver Hardware Setup	23
4.1.5	Received Data Observation	24
4.1.6	Complete Experimental Setup	25
4.2	Result Analysis	26
5	Conclusions and Future Scope	27
5.1	Conclusion	27
5.2	Future Scope	28
	References	28
	Appendix	30
A	Transmitter Program	30
B	Receiver Program	33
C	Hardware Connections	35
D	List of Acronyms	36

List of Tables

3.1	Hardware Specifications	16
3.2	Software Specifications	17
C.1	PYNQ-Z2 to NRF24L01 Connections	35

List of Figures

2.1	Functional Block Diagram of the Proposed System	11
3.1	Flowchart of the Proposed System	20
4.1	Vivado Block Design	22
4.2	Vitis UART Output	22
4.3	PYNQ-Z2 FPGA with NRF24L01 Transmitter	23
4.4	Arduino UNO with NRF24L01 Receiver	24
4.5	Arduino Serial Monitor Output	24
4.6	Complete Experimental Setup	25

Chapter 1

Introduction

1.1 Motivation

The fast-paced urbanization and the growing populations have caused a very sharp increase in the number of cars all over the world. Transportation systems have As a result been facing a range of issues like traffic jams, car accidents, ineffective traffic management, longer travelling time, etc. In fact, traditional transportation systems generally lack real-time information sharing feature. That means, their ability to respond to varying traffic conditions is rather restricted.

Integrating communication, computing and control technologies into the transportation infrastructures have made Intelligent Transportation Systems (ITS) a suitable answer to these problems. Effective sharing of information and Because of this taking actions based on these exchanges is majorly facilitated by the real-time communication which is also among the most vital parts of ITS operations.

With the help of Field Programmable Gate Arrays (FPGAs), the communication system can be made very flexible, powerful, and high-performing due to their parallel processing capability and real-time operation capability, respectively. The coupling of wireless communication modules with FPGA technology can greatly facilitate the production of extremely potent communication systems suitable for intelligent transportation environment use.

The principal objective of this thesis is to design FPGA-based Real-Time Communication System for ITS. Utilizing PYNQ-Z2 FPGA board and NRF24L01 wireless transceiver modules, the proposed communication system makes possible the real-time, wireless data transmissions between the transmitter and receiver nodes that can be extended to other ITS applications.

1.2 Intelligent Transportation Systems

Intelligent Transportation Systems or ITS aim at the utilization of information and telecommunication technologies to make travel safer, more comfortable, and more sustainable.

Technically, ITS merges the power of several technologies, like sensors, wireless communication, embedded systems, and big data analysis, to come up with smart ways to control transport.

Some of main the targets of ITS are:

Making the traffic environment safer. Reducing road congestion. Improving the performance of transportation.

Providing real-time traffic information. Monitoring and managing transport systems efficiently. Handling emergencies effectively.

Currently, the scope of ITS is so wide as to include vehicle tracking, traffic light control with artificial intelligence, electronic toll collection, and V2V (vehicle-to-vehicle) as well as V2I (vehicle-to-infrastructure) communications.

The very foundation of the achievement of ITS lays in the formation of super reliable communication systems that can exchange data among different parts of the transportation system in

real time for better coordination and, That's why, improvement of the overall transport network.

1.3 FPGA in Transportation Applications

FPGAs or Field Programmable Gate Arrays, thanks to their unique mixture of reprogrammability and very high computational speed, are determining changes for us in communication and transport applications.

Plans or algorithms are executed by microcontrollers one after another sequentially while FPGAs, Then again, can carry out various operations simultaneously through parallel processing.

Here is a list of a few of the advantages of using FPGA technology:

Blazing fast processing capability.

Extremely low communication overhead.

Achieving numerous tasks concurrently.

A reprogrammable hardware structure.

Changing and scaling up to meet new demands can be very easily done.

FPGA-driven solutions for Intelligent Transportation Systems may, for instance, take on communication processing, traffic surveillance, signal regulation, sensor interfacing, and data handling in real-time, and a whole lot of others. In transport, decision-making is frequently very rapid, and since FPGAs lead to several data streams being processed simultaneously, they become quite a logical choice for such situations.

We can use the PYNQ-Z2 FPGA as a versatile communication system and embedded solutions design platform.

1.4 Wireless Communication in Intelligent Transportation Systems

In fact, wireless communication makes a difference among the key technologies that ITS rely on. Various elements of the transport system like vehicles, roadside units, traffic surveillance stations, and control centers require constant data communication to offer smart transportation services. Wireless communication can be characterized with a few advantages:

Quite a bit reduced wiring.

Installation is possible from almost any location.

Systems can be expanded.

Real time data exchange.

Lower cost.

NRF24L01 wireless transceiver is a small module performing at the 2.4 GHz ISM band and is widely recognized for its solid short-range wireless communication capability. It is designed to operate in a packet communication mode and can be interfaced to FPGAs through an SPI interface. NRF24L01 wireless module is the core of the proposed system allowing FPGA-based transmitter to communicate wirelessly with Arduino-based receiver.

1.5 Objectives

The project aims to develop a real-time wireless communication system that can be used in Intelligent Transportation System (ITS) projects.

The specific objectives are:

- Understand the communication features required for an Intelligent Transportation System.
- Interface the NRF24L01 wireless module with the PYNQ-Z2 board.
- Set up SPI communication between the FPGA and the wireless module.
- Use NRF24L01 modules to exchange messages.
- Data acquisition through Arduino UNO.
- Demonstrate real-time communication of information between the communication nodes.
- Evaluate the applicability of FPGA-based communication systems in the transportation sector

1.6 Literature Survey

Advanced continuous communication networks, low latency, and real-time information sharing have augmented ITS.

Targets of FPGA communication systems are mainly their high processing speed and (re) configuration flexibility advantages amongst less exploited methods. Wireless communication is the fundamental technology to enable ITS applications.

Surveys show that users considered various communication modes such as Dedicated Short Range Communication (DSRC), Cellular Vehicle-to-Everything (C-V2X), 5G communication, and low-power wireless communication. The NRF24L01 transceiver, with its low power consumption, low cost, and easy integration, has become very popular and one of the most used devices for wireless sensor networks and embedded communication systems. The incorporation of wireless communication systems based on FPGA is a very promising area for real-time communication applications.

It turns out that the combination of FPGA and wireless communication technology, as the literature, is a very effective solution for Intelligent Transportation System applications.

1.7 Problem Statement

In current transportation systems, the biggest problem is the lack of communication means with good reliability that can handle real-time information exchange. Existing methods are not only very expensive to build the infrastructures but also inflexible and problematic during the deployment phase. The communication system carrying the benefits of FPGA technology and wireless communication can be one step to the efficient real-time data transfer.

This project makes this step by creating and first time adopting an FPGA-based wireless communication system with the help of PYNQ-Z2 FPGA board and NRF24L01 transceiver modules.

1.8 Organization of the Report

The report is structured into this chapters:

- **Chapter 1: Introduction**

This chapter gives an overview of the rationale and objectives of the investigation, outlines the literature review, defines the problem statement and presents the proposed communication system.

- **Chapter 2: System Design**

This chapter presents an overall description of the system architecture, detailed functional block diagrams, as well as separate descriptions of the transmitter and receiver components.

- **Chapter 3: Implementation Details**

This chapter deals with the physical and logical components of the system and a thorough explanation of the architecture, algorithm, and flowchart.

- **Chapter 4: Results and Discussions**

This chapter comprises of the experimental system description, implementation results, their interpretation and performance evaluation.

- **Chapter 5: Conclusions and Future Scope**

This final chapter summarizes the project findings and outlines potential future enhancements.

Chapter 2

System Design

2.1 Introduction

The designing of a system is an extremely essential step in the development of communication setups among other systems and technologies. It defines the system's architecture, operational flow, hardware components, and the communication methods that will be deployed in the new system.

The objective of the planned FPGA-Based Real-Time Communication Structure for Intelligent Transportation Systems is to facilitate reliable wireless communication between two communication nodes through the use of FPGAs and wireless transceiver modules.

The proposed system consists of the transmitter and the receiver. The transmitter is implemented using PYNQ-Z2 FPGA board with an NRF24L01 wireless transceiver module. The receiver consists of another NRF24L01 transceiver connected to an Arduino UNO. Communication between the transmitter and the receiver occurs through wireless transmission in the 2.4 GHz ISM band.

This system models the main point of real-time data communication within Intelligent Transportation Systems. The data communicated may be representative of traffic conditions, vehicle details, alert messages, or any transport-related information that gets exchanged between different transport parties.

2.2 Need for Real-Time Communication in Intelligent Transportation Systems

At the heart of Intelligent Transportation Systems is a communication technology component that makes easier the exchange of information between vehicles, roadside units, and central operation centers. The effectiveness of transportation systems depends on the ability to acquire, process, and distribute data on time.

Some of the advantages of real-time communication include:

- Mitigation of traffic congestion.
- Improvement of road safety.
- Faster response of emergency services.
- More efficient regulation of traffic lights.
- Enhanced vehicle tracking.
- Transport planning with more precision.

If packets are dropped and communication is delayed, the decisions can become flawed and the overall system performance can be affected. That means, low-latency communication networks are crucial for modern transportation applications.

2.3 Overall System Architecture

The communication system, which has been proposed, consists of two primary segments:

1. Transmitter Section
2. Receiver Section

At first, the transmitter portion captures the input data and After that processes the information utilizing the FPGA. The final step sends the data wirelessly with the help of the NRF24L01 transceiver. On the other side, the receiver section takes the transmitted packet and displays the received data on Arduino UNO.

The block diagram of the system shows how the different hardware modules work together in the proposed design.

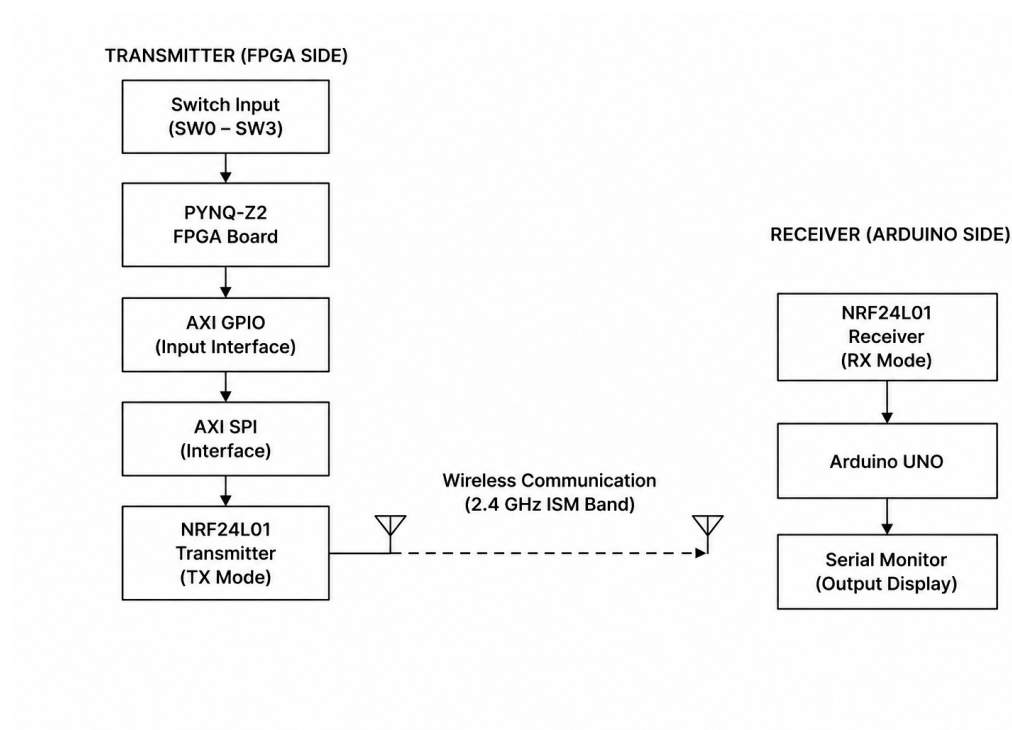


Figure 2.1: Functional Block Diagram of the Proposed System

The blocks in the system are ::

- Input Switches

- PYNQ-Z2 FPGA
- AXI GPIO
- AXI SPI
- NRF24L01 Transmitter
- Wireless Communication Channel
- NRF24L01 Receiver
- Arduino UNO
- Serial Monitor

2.4 Transmitter Section

The transmitter generates the information and sends it over the air.

The main parts of the transmitter are:

2.4.1 Input Switches

Input switches produce control information. The FPGA continuously tracks the switch condition via the AXI GPIO interface.

2.4.2 PYNQ-Z2 FPGA Board

The PYNQ-Z2 FPGA board is the heart of the transmitter. It gets the status of the switches, interprets the data, and triggers the wireless transmission.

FPGA can offer:

- Very fast processing
- Several instructions can be performed at the same time
- Capable of working in real-time
- Hardware can be configured as per requirements

2.4.3 AXI GPIO

The AXI GPIO core is used for connecting the switches to the FPGA processing system. It permits the processor to check the input signal level and create the matching communication packets.

2.4.4 AXI SPI

With the AXI SPI core, the FPGA interacts with the NRF24L01 wireless module. In the SPI mode of operation, a very fast synchronous information exchange is possible.

2.4.5 NRF24L01 Transmitter

The NRF24L01 unit is prepared in the transmitter function. It takes the data from the FPGA and sends out wireless packets in the 2.4 GHz ISM band.

2.5 Receiver Section

The receiver section picks up the wireless packets and shows the received data.

Those are the components of the receiver:

2.5.1 NRF24L01 Receiver

NRF24L01 receiver-board after receiving the wireless packet from the transmitter-passes communication data to Arduino UNO board

2.5.2 Arduino UNO

Arduino UNO is the main controller for the receiver. It takes the received data, processes it, and the output is shown on the serial monitor.

Reasons for choosing Arduino UNO:

- Easy to program
- Supports SPI
- Has built-in UART communication
- Cheap
- Allows for very fast prototyping

2.5.3 Serial Monitor

With the help of the Arduino Serial Monitor a simple and efficient interface can be provided to the user for viewing the data received over the wireless link. If a communication has been successful it means that transmitter and receiver nodes are properly communicating.

2.6 Wireless Communication Channel

The communication between the transmitter and receiver is established using NRF24L01 modules in the 2.4 GHz ISM band.

Using wireless communication channel brings with it some advantages, like:

- No physical wiring required
- Flexible deployment
- Low installation cost
- Real-time communication
- Easy scalability

NRF24L01 are able to support packet-based communication and they provide a reliable wireless connectivity that is ideal for a short-range application.

2.7 Communication Protocol

SPI protocol is used in the proposed system as a communication protocol between FPGA and the NRF24L01 module.

SPI uses these signals:

- MOSI (Master Out Slave In)
- MISO (Master In Slave Out)
- SCK (Serial Clock)
- CSN (Chip Select)

SPI is characterized by a very high rate of data transfer and low software overhead; Because of this, it is a suitable choice for real-time applications.

2.8 System Operation

The implementation of the communication system proposed here basically follows these steps from the system perspective:

1. Switch data is fetched from the AXI GPIO.
2. Pixel FPGA analyzes the input data.
3. Through SPI interface, the FPGA sends the processed data to the NRF24L01 module.
4. The NRF24L01 sends the data wirelessly.
5. The NRF24L01 receiver obtains the radio packet.
6. The obtained data gets equalized with the Arduino UNO.
7. Using the Serial Monitor, Arduino visualizes the incoming data.

2.9 Advantages of the Proposed System

Here are some notable benefits of the new communication structure:

- Real-time wireless communication.
- FPGA-based high-speed processing.
- Low-cost implementation.
- Flexible architecture.
- Easy scalability.
- Reduced wiring complexity.
- Suitable for Intelligent Transportation System applications.

2.10 Summary

This chapter has elaborated on the overall system of the FPGA-Based Real-Time Communication Setup for Intelligent Transportation System that we have proposed. We have dived into the system architecture, details of the transmitter, details of the receiver, communication protocol, and the operation flow of the system.

Chapter 3

Implementation Details

3.1 Specifications and System Architecture

The proposed FPGA-based real-time communication framework for Intelligent Transportation Systems consists of a transmitter section and a receiver section. The transmitter is implemented using the PYNQ-Z2 FPGA board interfaced with an NRF24L01 wireless transceiver module through the SPI protocol. The receiver consists of an Arduino UNO interfaced with another NRF24L01 transceiver module configured in receive mode.

The FPGA reads the status of the input switch and processes the corresponding data. The processed information is transmitted wirelessly using the NRF24L01 module operating in the 2.4 GHz ISM band. The receiver NRF24L01 module receives the transmitted data and forwards it to the Arduino UNO. The Arduino processes the received information and displays the corresponding output through the serial monitor.

The proposed system provides a simple and efficient communication framework suitable for Intelligent Transportation System applications where real-time wireless information exchange is required.

Hardware Specifications

Table 3.1: Hardware Specifications

Component	Specification
PYNQ-Z2	FPGA Board
NRF24L01	Wireless Transceiver Module
Arduino UNO	Receiver Controller
Communication Protocol	SPI
Operating Frequency	2.4 GHz ISM Band
Power Supply	3.3V / 5V

Software Specifications

Table 3.2: Software Specifications

Software Tool	Purpose
Vivado 2025.1	Hardware Design
Vitis 2025.1	Embedded Software Development
Arduino IDE	Receiver Programming
PuTTY	UART Monitoring
Embedded C	Application Development

System Architecture

The system architecture of the proposed FPGA-Based Real-Time Communication Structure for Intelligent Transportation Systems is illustrated in Figure ???. The architecture is divided into two primary components: the transmitter section and the receiver section.

In the transmitter section, a PYNQ-Z2 FPGA board is used which is connected to an NRF24L01 wireless transceiver module. The input source is a single onboard switch (SW0). The switch status is made available through the AXI GPIO interface and the FPGA processes it. The AXI SPI interface assists in the communication with the NRF24L01 module which wirelessly sends the data over the 2.4 GHz ISM band.

The receiver section incorporates another NRF24L01 transceiver module switched to receive mode and linked to an Arduino UNO. The receiver obtains the wireless packets sent from the other side and passes them on to the Arduino UNO. The Arduino interprets the received data and shows the corresponding output via the Serial Monitor.

The proposed architecture exhibits real-time wireless communication between the FPGA and Arduino platforms. This communication setup could be developed for Intelligent Transportation System applications including traffic monitoring, vehicle information exchange, and smart transportation infrastructure.

3.2 Algorithm

The operation of the proposed system is performed according to the following algorithm:

1. Start the system.
2. Initialize the PYNQ-Z2 FPGA platform.
3. Configure AXI GPIO for switch input.
4. Configure AXI SPI for communication with the NRF24L01 transmitter.
5. Initialize the NRF24L01 transceiver.
6. Read the switch status continuously.
7. Generate data corresponding to the switch state.
8. Transmit the data wirelessly through the NRF24L01 module.
9. Initialize the receiver NRF24L01 module connected to Arduino UNO.

10. Receive the transmitted wireless data.
11. Display the received information through the serial monitor.
12. Repeat the process continuously.
13. Stop the system.

3.3 PYNQ-Z2 FPGA Board

This PYNQ-Z2 FPGA development board is the heart of the proposed communication structure. Designed around a Xilinx Zynq-7000 System-on-Chip architecture, it couples a high-performance dual-core ARM Cortex-A9 processing system with Programmable Logic (PL) resources. This way you get both software and hardware implementations on the same platform. In the proposed system, the FPGA monitors the switch inputs, processes the generated information, and manages the wireless communication operation with the NRF24L01 transceiver module. The board was chosen due to its high interface processing speed, flexibility, and capability to support real-time applications. Thanks to its parallel processing capability, FPGA technology represents a good platform to build communication systems for use in Intelligent Transportation System environments.

3.4 NRF24L01 Wireless Transceiver

One way to establish communication between the transmitter section and the receiver section is to use the NRF24L01 wireless transceiver module. It makes use of the license-free 2.4 GHz ISM band and supports packet-based communication wirelessly. It communicates with the FPGA using the SPI protocol, which enables high-speed data transfer between the controller and the transceiver.

The NRF24L01 transmitter in this system gets data from the FPGA and sends it over the air. The other NRF24L01 in receive mode that has been set up is able to get the sent packet and give it to the Arduino UNO. The low power consumption, ease of interfacing, and strong communication features make the NRF24L01 the best fit for this project.

3.5 Arduino UNO

The Arduino UNO was used as the receiving controller in the communication setup proposed. It gets wireless packets through the NRF24L01 module and decodes the received data. The Arduino UNO which is built on the ATmega328 microcontroller supports SPI and UART communication protocols. The Arduino Serial Monitor was used to display the received data for checking the wireless communication. The choice of the board was based on the consideration of its easy programming environment, low price, and wide support for embedded communication applications.

3.6 Vivado Design Implementation

The hardware platform for the proposed system was created using Vivado Design Suite. At first, a new project was started and the ZYNQ7 Processing System was added. Then AXI GPIO and AXI Quad SPI peripherals were put into the block design to make the switch interfacing and communication with the NRF24L01 module easier. The design was then checked to make sure that the different hardware components were connected properly. When the check came

out fine, the hardware bitstream was produced and taken to Vitis for software development. The finished block design is the basis of the communication structure and it allows the processing system to work with the external hardware modules.

3.7 Flowchart

The flowchart of the proposed communication framework is shown in Figure 3.1.

The flowchart illustrates the sequence of operations performed by the transmitter and receiver sections. The FPGA continuously monitors the switch input and transmits the corresponding data through the NRF24L01 module. The receiver section receives the transmitted information and displays it through the serial monitor, thereby establishing real-time wireless communication.

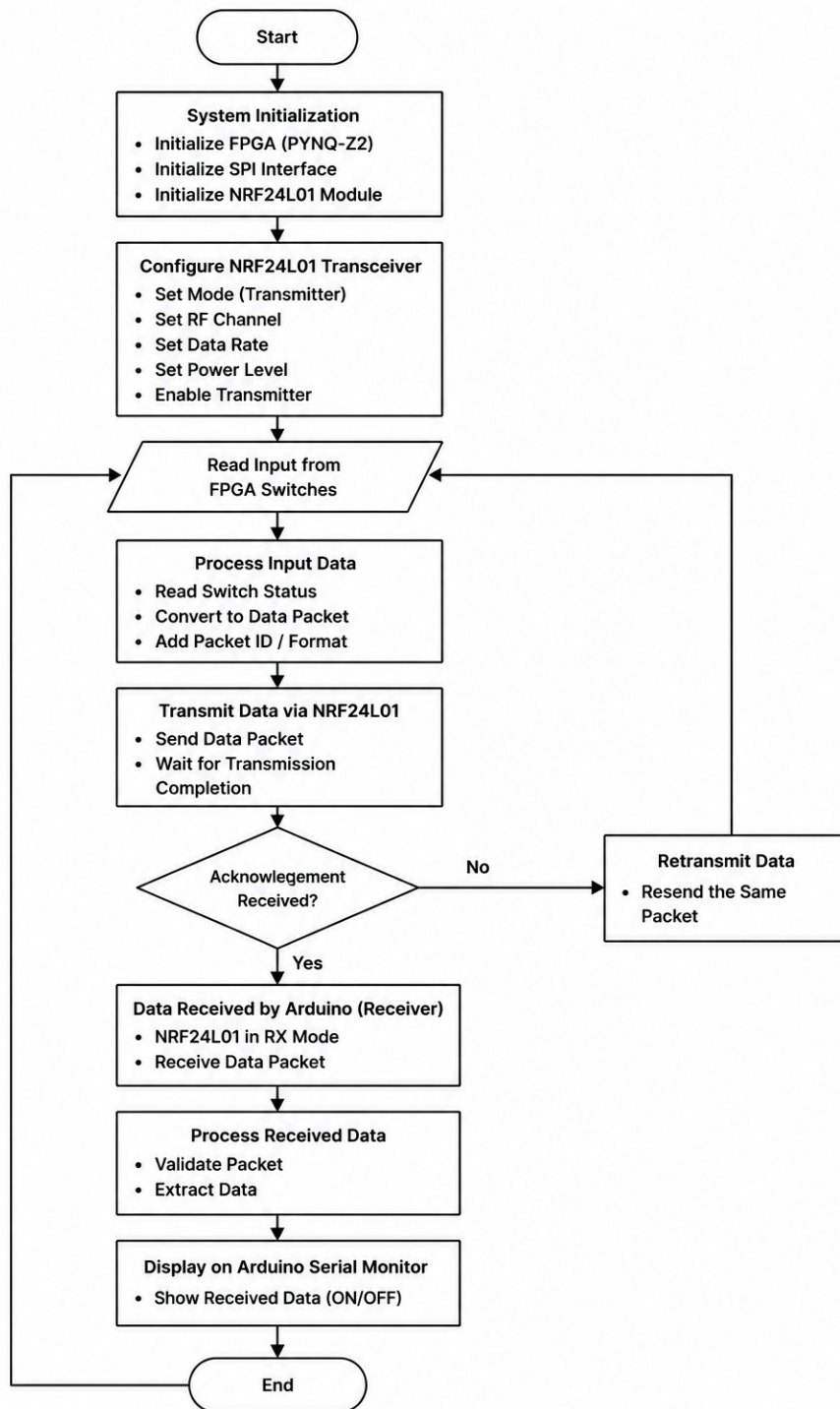


Figure 3.1: Flowchart of the Proposed System

Chapter 4

Results and Discussions

4.1 Results and Discussions

The proposed FPGA-based Real-Time Communication Framework for Intelligent Transportation Systems was successfully implemented using the PYNQ-Z2 FPGA board, NRF24L01 wireless transceiver modules, and Arduino UNO. The transmitter section was developed using the PYNQ-Z2 FPGA platform interfaced with the NRF24L01 transceiver through the SPI protocol. The receiver section consisted of an Arduino UNO connected to an NRF24L01 receiver module.

The FPGA successfully transmitted data through the NRF24L01 wireless transceiver operating in the 2.4 GHz ISM band. The receiver module received the transmitted information and displayed the corresponding data through the Arduino Serial Monitor. The communication process demonstrated reliable wireless transmission and reception suitable for Intelligent Transportation System applications.

The implementation process involved hardware design using Vivado, software development using Vitis, wireless communication through NRF24L01 modules, and data reception using Arduino UNO. The obtained results validate the successful operation of the proposed communication framework.

4.1.1 Vivado Block Design

The hardware architecture was designed using Vivado. The design consists of the ZYNQ7 Processing System, AXI GPIO for switch interfacing, and AXI SPI for NRF24L01 communication.

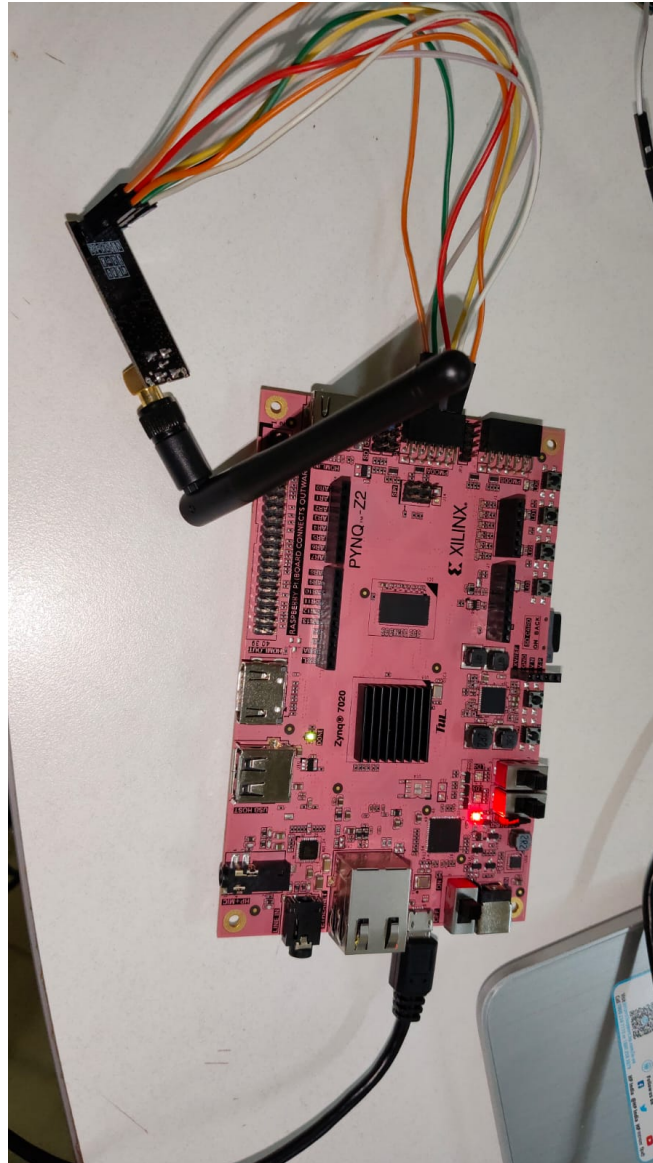


Figure 4.3: PYNQ-Z2 FPGA with NRF24L01 Transmitter

4.1.4 Receiver Hardware Setup

The receiver section consists of an Arduino UNO connected to an NRF24L01 transceiver configured in receive mode. The receiver continuously listens for incoming wireless packets.

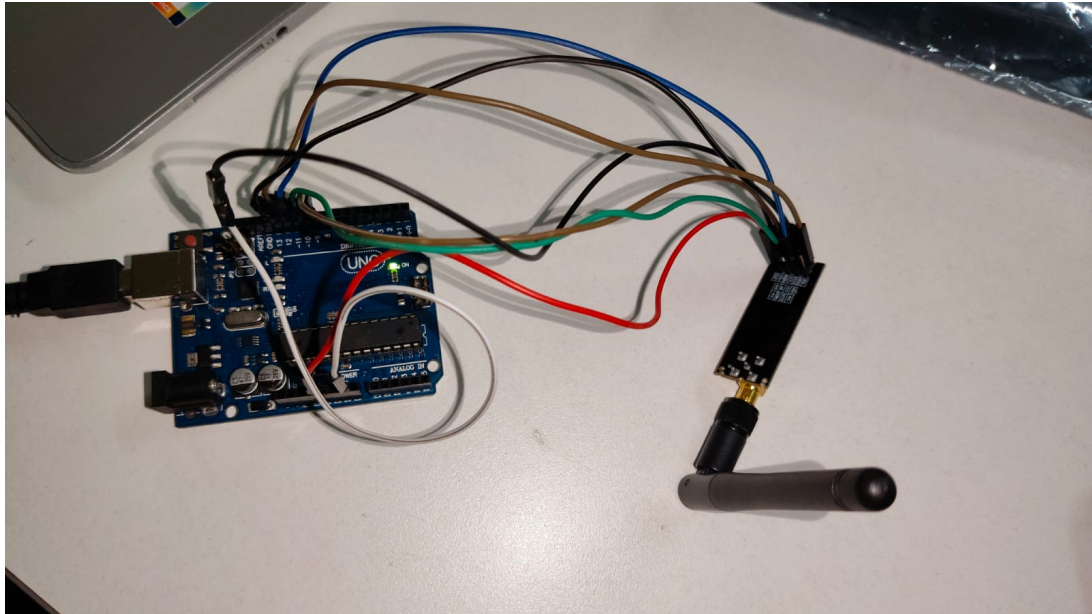


Figure 4.4: Arduino UNO with NRF24L01 Receiver

4.1.5 Received Data Observation

The transmitted information was successfully received and displayed on the Arduino Serial Monitor. The received data matched the transmitted switch status, validating successful wireless communication.

```

44 {
45   if (radio.available())
46   {
47     radio.read(&receivedData, sizeof(receivedData));
48     if (receivedData == '0')
49     {
50       Serial.println("Received = 0");
51     }
52     else if (receivedData == '1')
53     {
54       Serial.println("Received = 1");
55     }
56     else
57     {
58       Serial.print("Unknown Data = ");
59       Serial.println(receivedData);
60     }
61     delay(1000); // 1 second delay
62   }
63 }
64 }

```

Output Serial Monitor x

Message (Enter to send message to 'Arduino Uno' on 'COM5')

Received = 0
 Received = 1
 Received = 0
 Received = 1
 Received = 0
 Received = 1
 Received = 0
 Received = 1
 Received = 0
 Received = 1

Ln 64, Col 2 Arduino Uno on COM5

Figure 4.5: Arduino Serial Monitor Output

4.1.6 Complete Experimental Setup

The complete hardware implementation of the proposed communication framework is shown below. The setup consists of the PYNQ-Z2 FPGA transmitter, NRF24L01 wireless modules, and Arduino UNO receiver.

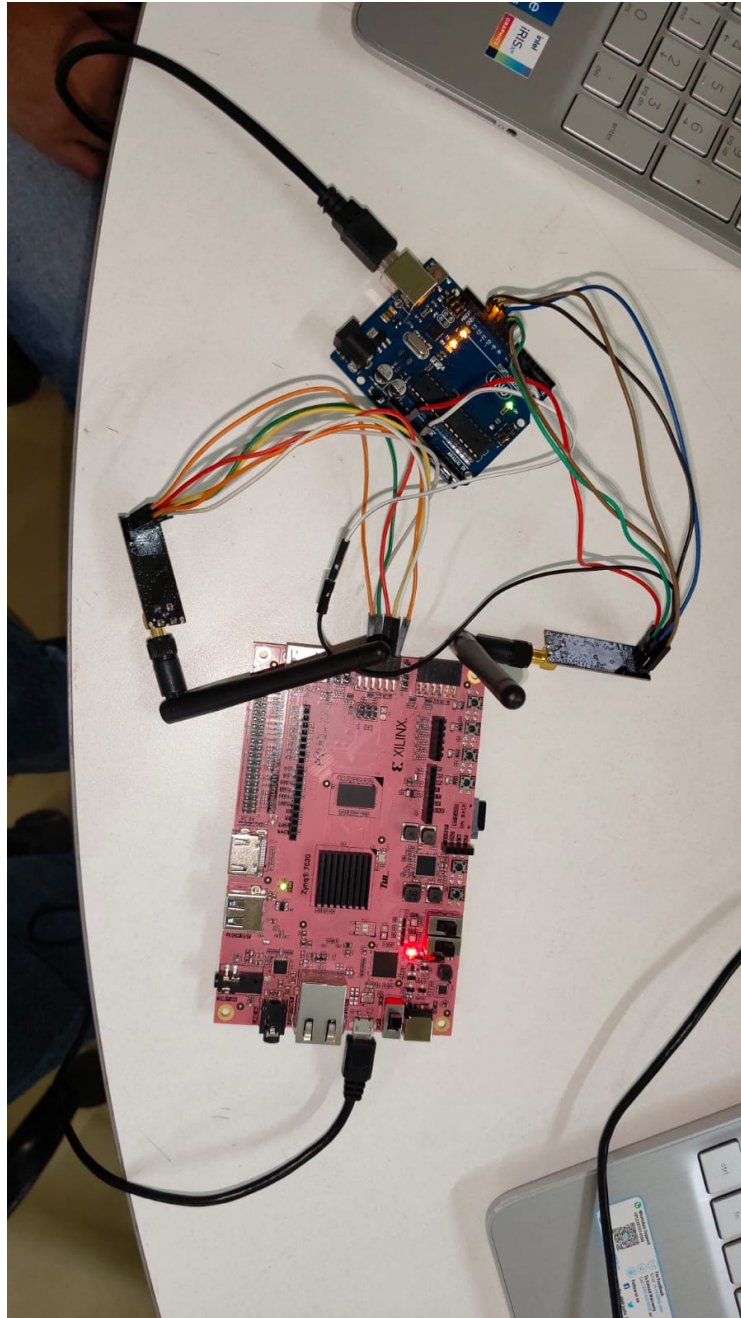


Figure 4.6: Complete Experimental Setup

4.2 Result Analysis

The experimental results confirm the successful implementation of the FPGA-based wireless communication framework. The PYNQ-Z2 FPGA successfully interfaced with the NRF24L01 module through the SPI protocol and transmitted data wirelessly. The receiver NRF24L01 module connected to the Arduino UNO successfully received the transmitted packets and displayed the received information through the Serial Monitor.

The following observations were obtained during experimentation:

- Successful interfacing of the NRF24L01 transceiver with the PYNQ-Z2 FPGA board.
- Successful implementation of SPI communication between FPGA and NRF24L01.
- Reliable wireless communication in the 2.4 GHz ISM band.
- Successful reception of transmitted data using Arduino UNO.
- Real-time display of received information through the Serial Monitor.
- Stable communication performance under normal operating conditions.
- Verification of wireless communication suitable for Intelligent Transportation System applications.

The obtained results demonstrate that the proposed communication framework provides an effective platform for real-time wireless information exchange. The combination of FPGA technology and wireless communication enables efficient and scalable communication suitable for future Intelligent Transportation System applications such as traffic monitoring, smart transportation infrastructure, and vehicle communication systems.

Chapter 5

Conclusions and Future Scope

5.1 Conclusion

The objective of this project was to design and implement an FPGA-Based Real-Time Communication Framework for Intelligent Transportation Systems using the PYNQ-Z2 FPGA board, NRF24L01 wireless transceiver modules, and Arduino UNO. The increasing demand for efficient transportation systems requires reliable communication frameworks capable of supporting real-time information exchange. In this context, the proposed system demonstrates a simple and cost-effective wireless communication framework suitable for transportation-related applications.

The transmitter section was implemented using the PYNQ-Z2 FPGA platform interfaced with the NRF24L01 wireless transceiver through the SPI protocol. The FPGA successfully controlled the communication process and transmitted data wirelessly through the 2.4 GHz ISM band. The receiver section consisted of an Arduino UNO connected to another NRF24L01 module configured in receive mode. The received information was processed by the Arduino and displayed through the Serial Monitor.

The implementation involved hardware design using Vivado, embedded software development using Vitis, and receiver programming using Arduino IDE. The successful integration of these components demonstrated the capability of FPGA technology to support wireless communication systems. The communication process was observed to be reliable under normal operating conditions, and the transmitted information was successfully received at the receiver side.

The project provided practical exposure to FPGA design, wireless communication protocols, SPI interfacing, embedded software development, and system integration. The use of FPGA technology offered advantages such as high processing speed, parallel execution capability, flexibility, and scalability. These characteristics make FPGA-based communication systems suitable for Intelligent Transportation System applications where real-time operation is essential.

The experimental results validated the successful operation of the proposed communication framework. The transmitter and receiver modules established wireless communication effectively, and the received data accurately reflected the transmitted information. The system demonstrated the feasibility of using FPGA-based wireless communication frameworks for transportation-related applications requiring real-time information exchange.

Overall, the proposed system achieved all the objectives of the project and established a reliable wireless communication link between the FPGA-based transmitter and the Arduino-based receiver. The developed framework serves as a foundation for future research and development in Intelligent Transportation Systems and smart mobility applications.

5.2 Future Scope

The proposed communication framework can be further enhanced to support advanced Intelligent Transportation System applications. Future developments may focus on improving communication range, reliability, and scalability for real-world transportation environments.

The system can be extended to support Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communication, enabling efficient information exchange between vehicles and roadside units. Integration of sensors and GPS modules can facilitate real-time vehicle monitoring, tracking, and traffic management.

Security features such as encryption and authentication can be incorporated to ensure secure communication. Furthermore, the framework can be integrated with Internet of Things (IoT) platforms for cloud-based monitoring and data analysis. The application of Artificial Intelligence and Machine Learning techniques can also support intelligent traffic prediction and decision-making.

These enhancements can transform the proposed framework into a more robust and scalable communication system suitable for future Intelligent Transportation Systems and smart city applications.

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Appendix A

Transmitter Program

The following Embedded C program was developed in Vitis to configure the NRF24L01 module and perform wireless data transmission.

```
#include "xspi.h"
#include "xgpio.h"
#include "xparameters.h"
#include "xil_printf.h"

XSpi Spi;
XGpio SW_GPIO;

/* NRF Write Register */
void NRF_WriteReg(u8 reg, u8 value)
{
    u8 TxBuf[2];
    u8 RxBuf[2];

    TxBuf[0] = 0x20 | reg;
    TxBuf[1] = value;

    XSpi_Transfer(&Spi, TxBuf, RxBuf, 2);
}

/* NRF Send Payload */
void NRF_SendByte(u8 data)
{
    u8 TxBuf[2];
    u8 RxBuf[2];

    TxBuf[0] = 0xA0;          // W_TX_PAYLOAD
    TxBuf[1] = data;

    XSpi_Transfer(&Spi, TxBuf, RxBuf, 2);
}

int main()
{
    int Status;
```

```

u32 sw;
u32 prev_sw = 2;

xil_printf("\r\n");
xil_printf("==== NRF24 TX DEMO ==== \r\n\r\n");

/* SPI Init */
Status = XSpi_Initialize(&Spi, XPAR_XSPI_0_DEVICE_ID);

if(Status != XST_SUCCESS)
{
    xil_printf("SPI Init Failed\r\n");
    while(1);
}

XSpi_SetOptions(&Spi,
                XSP_MASTER_OPTION |
                XSP_MANUAL_SSELECT_OPTION);

XSpi_SetSlaveSelect(&Spi, 0x01);

XSpi_Start(&Spi);
XSpi_IntrGlobalDisable(&Spi);

xil_printf("SPI Ready\r\n");

/* NRF Configuration */
NRF_WriteReg(0x00, 0x0A);    // CONFIG
NRF_WriteReg(0x01, 0x00);    // EN_AA OFF
NRF_WriteReg(0x02, 0x01);    // EN_RXADDR
NRF_WriteReg(0x05, 40);      // RF_CH = 2440 MHz
NRF_WriteReg(0x06, 0x07);    // RF_SETUP

xil_printf("NRF Ready\r\n");

/* Switch GPIO */
Status = XGpio_Initialize(&SW_GPIO,
                          XPAR_AXI_GPIO_0_DEVICE_ID);

if(Status != XST_SUCCESS)
{
    xil_printf("SW GPIO Error\r\n");
    while(1);
}

XGpio_SetDataDirection(&SW_GPIO, 1, 1);

xil_printf("SW GPIO Ready\r\n\r\n");

while(1)

```

```

{
    sw = XGpio_DiscreteRead(&SW_GPIO, 1);

    if(sw != prev_sw)
    {
        prev_sw = sw;

        if(sw == 0)
        {
            NRF_SendByte('0');

            xil_printf("SW=0 -> Transmitting Data = 0\r\n");
            xil_printf("Packet Sent : 0\r\n\r\n");
        }
        else
        {
            NRF_SendByte('1');

            xil_printf("SW=1 -> Transmitting Data = 1\r\n");
            xil_printf("Packet Sent : 1\r\n\r\n");
        }
    }

    for(volatile int i=0;i<1000000;i++);
}

return 0;
}

```

Appendix B

Receiver Program

The following Arduino program was used to receive wireless packets through the NRF24L01 module.

```
#include <SPI.h>
#include <nRF24L01.h>
#include <RF24.h>

RF24 radio(9, 10);

const byte address[6] = "00001";

char receivedData;

void setup()
{
    Serial.begin(9600);

    if (!radio.begin())
    {
        Serial.println("NRF Failed");
        while (1);
    }

    radio.setChannel(40);           // 2440 MHz
    radio.setDataRate(RF24_1MBPS);
    radio.setAutoAck(false);

    radio.openReadingPipe(0, address);

    radio.startListening();

    Serial.println("Receiver Ready");
    Serial.println();
}

void loop()
{
    if (radio.available())
```

```
{
  radio.read(&receivedData, sizeof(receivedData));

  if (receivedData == '0')
  {
    Serial.println("Received = 0");
  }
  else if (receivedData == '1')
  {
    Serial.println("Received = 1");
  }
  else
  {
    Serial.print("Unknown Data = ");
    Serial.println(receivedData);
  }

  delay(1000);    // 1 second delay
}
}
```

Appendix C

Hardware Connections

NRF24L01 Pin	PYNQ-Z2 Pin
VCC	3.3V
GND	GND
MOSI	SPI MOSI
MISO	SPI MISO
SCK	SPI SCK
CSN	SPI SS
CE	GPIO

Table C.1: PYNQ-Z2 to NRF24L01 Connections

Appendix D

List of Acronyms

FPGA	Field Programmable Gate Array
ITS	Intelligent Transportation System
SPI	Serial Peripheral Interface
UART	Universal Asynchronous Receiver Transmitter
V2V	Vehicle-to-Vehicle
V2I	Vehicle-to-Infrastructure
RF	Radio Frequency
ISM	Industrial Scientific Medical